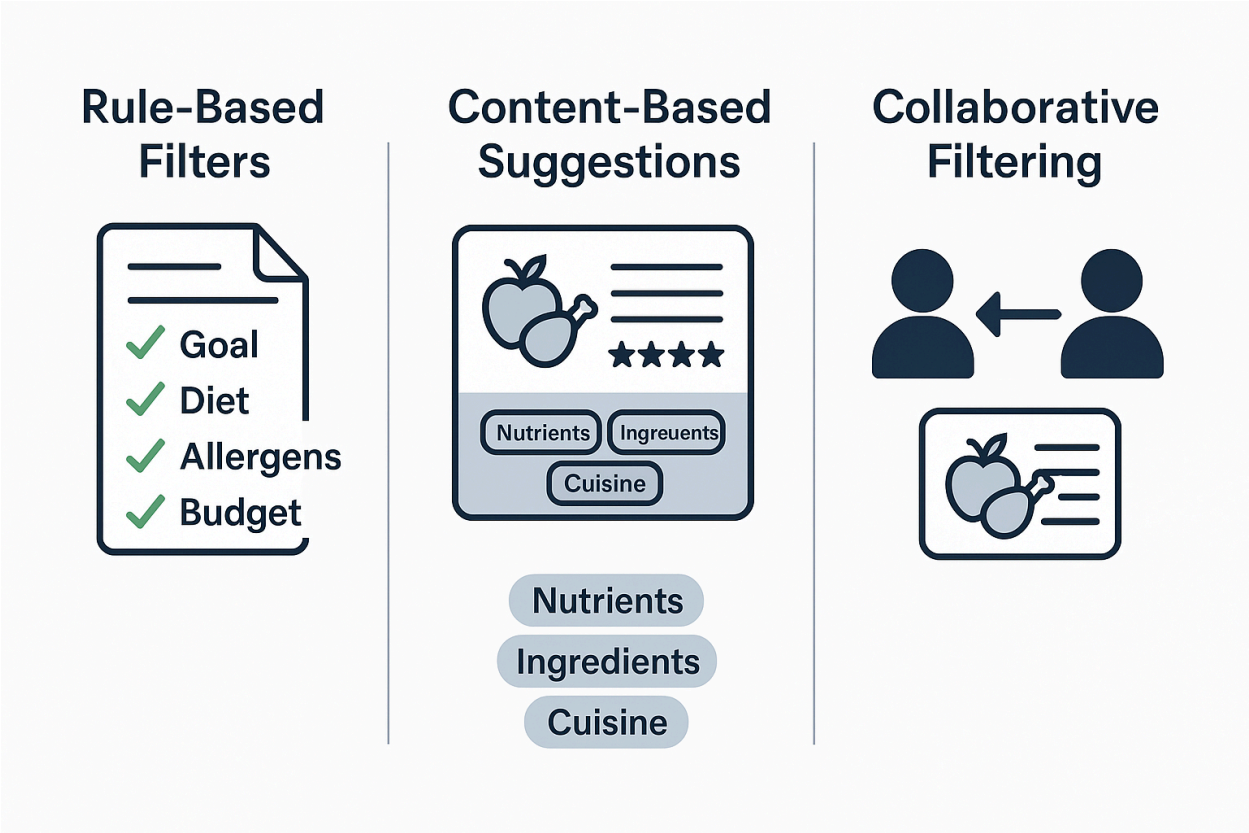
RE063 - Personalisation Algorithm Research

Introduction

For Food Remedy to actually feel useful, it needs to go beyond showing a list of meals and instead deliver recommendations that match the user's goals, preferences, and restrictions. Personalisation is what makes users feel like the app is made for them, not just a generic recipe database. In this research, I explored three common methods of personalising recommendations: **rule-based filters**, **content-based suggestions**, and **collaborative filtering**. I also looked at how popular nutrition and fitness apps currently handle personalisation to see what would work best for our MVP.



Methods of Personalisation

1. Rule-Based Filters

Rule-based systems apply the user's profile settings directly to the data. This means recommendations are shaped by specific criteria like diet type, allergens, disliked ingredients, prep time, or cost.

Advantages

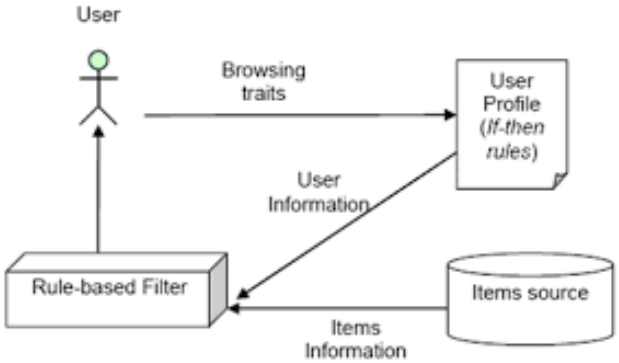
- Easy to implement with our stack.
- Completely explainable to the user ("we removed peanuts because you marked an allergy").
- Matches what users expect when they set diet preferences.

Disadvantages

- Can feel repetitive, since users with the same profile settings will see similar meals.
- Doesn't learn much from ongoing behaviour beyond the initial profile.

Examples in apps

- **Lifesum** uses plans like keto, paleo, or vegetarian, then shows recipes that fit those plans.
- **WeightWatchers** tailors meal suggestions to your personalised "points budget," which is still rule-driven at the core.



2. Content-Based Suggestions

Content-based methods look at the actual features of meals and try to match them with the tastes and goals of the user. Every meal needs to be tagged with details such as:

- Nutrients (calories, protein, sugar, fibre, sodium, fat)

- Ingredients
- Cuisine
- Prep time
- Cost band
- Meal type (breakfast, lunch, dinner, snack)

The system then learns from what the user interacts with. For example, if a user likes quick vegetarian recipes with high protein, it will recommend more meals with those traits.

Advantages

- Works from day one without needing lots of user data.
- Adapts quickly as the user saves, likes, or hides recipes.
- Can highlight variety by looking at multiple traits (nutrients, cuisine, prep time).

Disadvantages

- Relies on having well-tagged data, which can take effort to set up.
- If tagging is poor or inconsistent, suggestions won't feel accurate.

Examples in apps

- **Cronometer** goes deep on nutrients, so suggestions are highly aligned with personal nutrition goals.
- **MyFitnessPal** recommends meals that fit into your calorie and macro targets.

3. Collaborative Filtering

Collaborative filtering works by finding users with similar behaviour and recommending items they liked. In a food context, this could be: "users who saved this recipe also saved these others."

Advantages

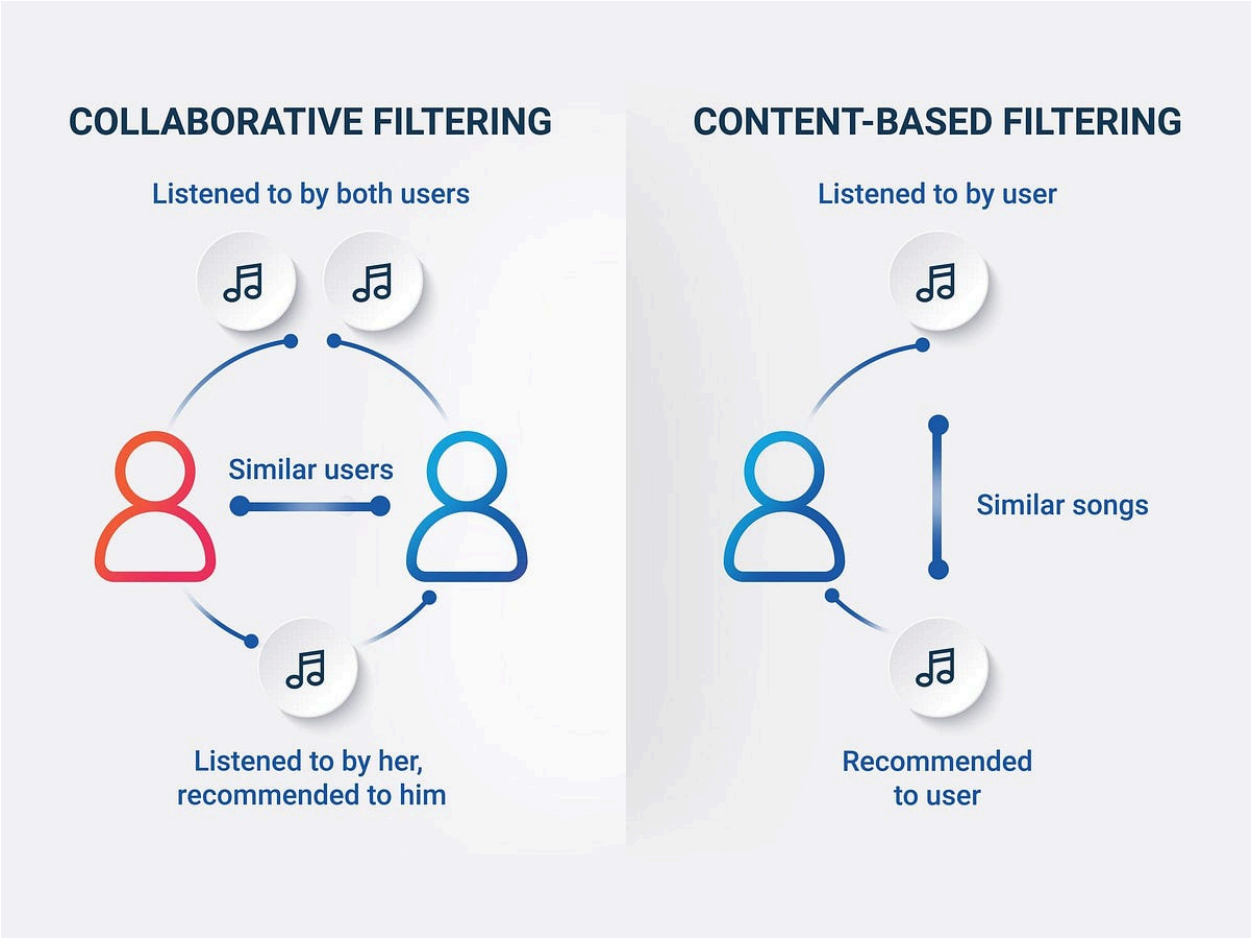
- Can uncover surprising and interesting suggestions that rules or tags might miss.
- Improves over time as more users interact with the app.

Disadvantages

- Suffers from the “cold start” problem, meaning it doesn’t work well without lots of data.
- More complex to implement and maintain compared to rules or content-based methods.

Examples in apps

- Streaming services like Netflix and Spotify are famous for collaborative filtering, but nutrition apps use it less often unless they have very large user bases.



What Other Apps Are Doing

Looking at well-known nutrition and fitness apps shows a clear trend:

- **Lifesum** and **WeightWatchers** mainly use rule-based filters with goal-driven plans.
- **MyFitnessPal** combines rules (macro goals) with content-based suggestions (meal logging + matching).
- **Cronometer** relies heavily on content-based approaches with nutrient-rich tagging.
- **Noom** simplifies personalisation with a colour system based on calorie density, which is basically a form of content-based scoring.

These apps show that hybrid approaches are common, usually starting with rules, then layering content-based or collaborative methods later as the user base grows.

Best Fit for Food Remedy MVP…

For our MVP, the best option is a **lightweight hybrid approach**:

- **Use rule-based filters** as hard constraints so meals always respect diet type, allergens, dislikes, prep time, and cost.
- **Add content-based suggestions** on top to rank and personalise results based on nutrients, cuisines, and user behaviour.

This combination is:

- **Lightweight**, we can implement it quickly without heavy machine learning.
- **Explainable**, users will understand why meals are recommended.
- **Adaptable**, it gets better as more interactions are logged.

Collaborative filtering can be introduced later once Food Remedy has enough users and saved recipes to make it worthwhile.

Conclusion

Food Remedy should begin with a hybrid of rule-based filters and content-based suggestions. This matches what successful apps are already doing and gives us the right balance of simplicity and personalisation. It ensures recommendations are safe and relevant from day one while allowing the system to gradually learn from user behaviour. Collaborative filtering can be planned as a future upgrade when the app has more scale.

This approach will let Food Remedy deliver personal meal recommendations that feel meaningful to the user, while staying lightweight enough to ship in our MVP timeline.